Notes: Buera and Shin (JPE, 2013)

Financial Frictions and the Persistence of History: A Quantitative Exploration

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1 Big picture

- **Question.** Why do the "economic miracle" episodes feature slow but sustained growth in output and TFP, rising investment rates, and gradual financial deepening, while the standard neoclassical transition predicts fast convergence and declining investment rates?
- **Core mechanism.** A reform removes idiosyncratic distortions that had misallocated resources across entrepreneurs. Reallocation should raise TFP quickly, but collateral constraints slow entry and expansion by productive but initially poor entrepreneurs.
- **State variable that creates persistence.** Wealth distribution matters. Productive entrepreneurs need collateral to operate at efficient scale. If productive entrepreneurs begin poor, they must accumulate wealth before reallocating resources fully.
- **Main quantitative result.** With financial frictions, the transition speed is about half that of the standard neoclassical benchmark. Investment rates are initially low and rise over time, matching the miracle-economy pattern.
- **TFP result.** TFP is endogenous. It rises because reform and gradual wealth accumulation reallocate capital and labor toward more productive entrepreneurs.
- **Historical persistence result.** The model makes history persistent: the initial distribution of wealth and entrepreneurial ability affects the transition path long after distortions are removed.
- **Evidence.** Data from China, Japan, Korea, Malaysia, Singapore, Taiwan, and Thailand show slow transitions, rising investment rates, rising TFP, gradual financial development, labor reallocation, rising private-sector shares, and changes in establishment size.
- **Economic takeaway.** Financial frictions can turn a one-time reform into a long development episode because efficient producers must first accumulate internal funds before expanding.

2 Incremental contribution of the paper

1. **From exogenous TFP to endogenous TFP dynamics.** Standard growth accounting treats TFP as exogenous. This paper generates TFP dynamics from reallocation across heterogeneous entrepreneurs subject to financial constraints.
2. **From steady-state misallocation to transition dynamics.** Prior work such as Restuccia–Rogerson and Hsieh–Klenow emphasizes static productivity losses from wedges. Buera and Shin ask how economies move after wedges are removed.
3. **A quantitative explanation of miracle-economy time paths.** The model jointly matches slow output convergence, initially rising investment rates, rising TFP, and late financial deepening, which the neoclassical model cannot match.
4. **Wealth distribution as macro state.** The paper shows that the distribution of wealth by entrepreneurial productivity is macroeconomically important because collateral constraints bind differently across entrepreneurs.
5. **Financial frictions as persistence technology.** Financial frictions do not merely lower steady-state output. They propagate historical distortions into the future by slowing the reallocation of capital and entry.
6. **Micro-to-macro bridge.** The paper links aggregate miracle dynamics to micro implications: entry/exit, average entrepreneurial ability, wealth concentration among high-ability entrepreneurs, labor reallocation, and establishment size.
7. **Policy implication.** Removing distortions is not enough. Without financial development, reforms can unlock productivity only gradually because the productive poor remain constrained.

Interpretation: The incremental contribution is the synthesis: a development episode is modeled as the interaction of reform, misallocation, entrepreneurial heterogeneity, and collateral-constrained wealth accumulation.

3 Data and measurement

3.1 Economic miracle sample

The empirical motivation uses seven miracle economies: China, Japan, Korea, Malaysia, Singapore, Taiwan, and Thailand. For each country, year 0 is the date of a major large-scale reform that marks the beginning of the transition.

- Output is per-worker GDP relative to the United States.
- Investment rate is the investment-to-output ratio.
- TFP is constructed relative to the U.S. and net of human capital contributions.
- Financial development is measured by private credit relative to GDP.
- Micro-level transition facts include labor reallocation, private-sector share, and average establishment size.

Interpretation: The empirical targets are transition shapes, not only levels. The model is evaluated by whether it generates rising investment rates, slow capital accumulation, and endogenous TFP growth over two decades.

3.2 Calibration targets

The calibration is disciplined by U.S. cross-sectional data and macro moments. Table 1 targets:

- employment share of the top 10 percent of establishments;
- earnings share of the top 5 percent;
- establishment exit rate;
- real interest rate.

The financial-frictions parameter λ is chosen to match a low external finance-to-GDP ratio, while the idiosyncratic distortions are chosen to generate the observed prereform misallocation and postreform transition dynamics.

Interpretation: The calibration combines micro concentration moments and macro financial-development moments. This is essential because the paper's mechanism is about how micro heterogeneity aggregates under financial constraints.

3.3 Model distance to data

The paper compares model transitions with the average miracle-economy transition using relative root mean squared error. Table 2 and Table 3 report distances relative to a perfect-credit transition.

Interpretation: The goal is not to match every country exactly. The goal is to show that introducing financial frictions moves the model significantly closer to the average miracle-economy transition, especially in the first five years.

4 Model and theoretical specification

4.1 Agents and occupational choice

Individuals differ in entrepreneurial productivity e and wealth a. Each period, an individual chooses whether to become a worker or an entrepreneur. Entrepreneurs operate a decreasing-returns technology, while workers supply labor for a wage.

$$y = zek^{\alpha}\ell^{\theta}, \qquad \alpha + \theta < 1. \tag{1}$$

Interpretation: Diminishing returns make managerial talent scarce. High-e entrepreneurs should run large firms, but financial constraints may prevent them from doing so.

4.2 Financial friction

Entrepreneurs face collateral constraints caused by imperfect contract enforcement. A simple representation is:

$$k \leq \lambda a, \tag{2}$$

where k is capital used in production, a is wealth, and λ measures the strength of financial development. A larger λ means weaker financial frictions.

Interpretation: The constraint links wealth to productive scale. If talented entrepreneurs are poor, they cannot immediately operate efficiently after reform.

4.3 Idiosyncratic distortions and reform

Before reform, idiosyncratic wedges distort entrepreneurial choices. Some entrepreneurs are subsidized and operate too much; others are taxed and operate too little. Reform removes these wedges at year 0.

$$\tau_i \neq 0 \quad \text{before reform}, \qquad \tau_i = 0 \quad \text{after reform}. \tag{3}$$

Interpretation: Reform eliminates the policy distortions, but it does not instantly eliminate the wealth distribution created by those distortions. This is the source of persistent history.

4.4 Reallocation margins

The model has two reallocation margins:

- **Intensive margin:** existing entrepreneurs adjust the scale of operations.
- **Extensive margin:** previously excluded high-productivity individuals enter entrepreneurship, while subsidized low-productivity entrepreneurs exit.

Interpretation: The intensive margin can move quickly. The extensive margin is slower because productive entrants must accumulate collateral before entering and growing.

4.5 Aggregate TFP

Aggregate TFP is endogenous because capital and labor are allocated across heterogeneous entrepreneurs:

$$TFP_t = F(\text{allocation of } k, \ell \text{ across } e \text{ types}). \tag{4}$$

Interpretation: TFP rises along the transition because inputs move from less productive to more productive entrepreneurs. Financial frictions slow that movement.

5 Identification and empirical design

This is a quantitative macro paper rather than a reduced-form causal design. Its identification is structural and disciplined by transition facts.

5.1 What the model identifies

The model isolates the role of financial frictions by comparing transitions after the same reform under different values of λ:

- $\lambda = \infty$ gives perfect credit and fast neoclassical-style adjustment.
- finite λ makes wealth distribution and collateral constraints matter.

Interpretation: The core counterfactual is: what would the same reform have done if entrepreneurs could borrow freely? The difference between that path and the financial-frictions path identifies the persistence created by limited finance.

5.2 Why initial wealth distribution matters

The prereform economy creates a distorted joint distribution of productivity and wealth. Some incompetent entrepreneurs accumulate wealth because they are subsidized. Some talented entrepreneurs remain poor because they are taxed or excluded.

$$(e, a)_0 \quad \text{summarizes historical misallocation.} \tag{5}$$

Interpretation: The paper's notion of history is concrete: history is the distribution of wealth across productivity types at the moment reform begins.

5.3 Distinguishing financial frictions from technology shocks

The paper compares the benchmark reform transition with a permanent aggregate technology shock. A pure technology shock produces fast adjustment and an investment-rate path that declines from a high initial level. The data instead show slow adjustment and rising investment rates.

Interpretation: This comparison shows why the miracle-economy facts are not well explained by standard neoclassical shocks. The distinctive shape of the investment rate is evidence for the reallocation-and-finance mechanism.

5.4 External validation with micro facts

The model predicts labor reallocation, growth of the private sector, and changes in establishment size. The paper compiles data on these margins from miracle economies.

Interpretation: These facts are not formal causal tests, but they increase the credibility of the mechanism because the model matches more than aggregate output and investment.

5.5 Limitations

- Reform timing is stylized as a discrete removal of distortions.
- The idiosyncratic wedges are reduced-form proxies for complex policy distortions.
- Financial frictions are summarized by one collateral parameter.
- The model abstracts from many open-economy channels, human-capital dynamics, and sector-specific policy details.

Interpretation: The strength of the paper is not institutional detail for any one country, but a quantitative mechanism that matches broad transition patterns across miracle economies.

6 Figures and tables

The following visuals are directly cropped from the paper. Adjacent related figures and tables are paired on the same row whenever readable. Single visuals are set to width 0.6 as requested.

6.1 Figure 1: Transitional dynamics from the economic miracles; Figure 2: Long-run effect of financial frictions

Read: Figure 1 shows miracle-economy transitions in GDP, investment, TFP, and private credit. Figure 2 shows how stronger financial frictions lower long-run GDP and TFP and depress the interest rate.

Economic message: Figure 1 presents the empirical puzzle. Figure 2 shows the model's basic steady-state mechanism: tighter finance misallocates resources and lowers aggregate productivity.

Detailed interpretation: The first figure shows three facts the model must match: output and TFP rise gradually, investment rates rise initially, and financial development comes late. The second figure explains why financial constraints can reduce both static efficiency and dynamic adjustment.

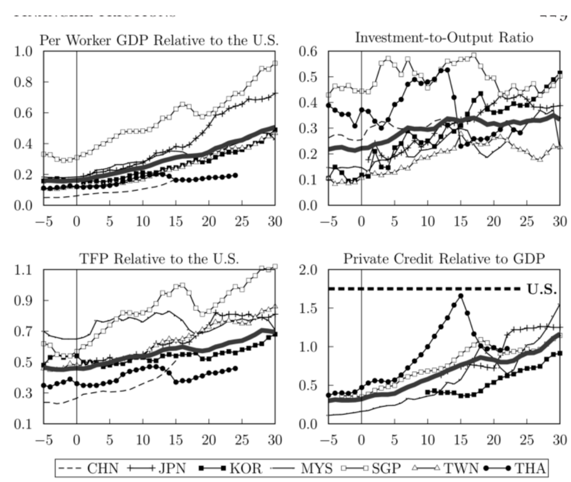


FIG. 1.—Transitional dynamics from the economic miracles. In each panel, all available

(a) Figure 1: transitional dynamics from the economic miracles.

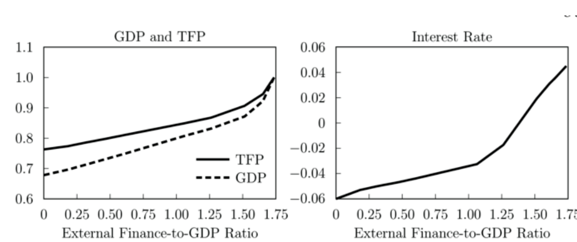


FIG. 2.—Long-run effect of financial frictions. The horizontal axes show the model-implied long-run external finance-to-GDP ratio for a given λ. The GDP and TFP series on the left panel are normalized by their respective values in the perfect credit economy

(b) Figure 2: long-run effect of financial frictions.

6.2 Table 1: Calibration; Table 2: Model distance to data, relative to perfect credit transition

Read: Table 1 reports the targeted U.S. moments and calibrated parameters. Table 2 reports how much closer the benchmark financial-frictions model is to the data relative to a perfect-credit transition.

Economic message: The calibration pins down heterogeneity, span of control, entrepreneurial dynamics, and discounting. The distance table shows the quantitative payoff of introducing financial frictions.

Detailed interpretation: Table 2 is especially important because it turns visual fit into a number. The model with financial frictions reduces the distance to output and investment-rate data substantially in the first five years and over 20 years.

TABLE 1
CALIBRATION

	US Data (1)	Model (2)	Parameter (3)
Top 10% employment	.67	.67	$\eta = 4.15$, $\nu = .21$
Top 5% earnings	.30	.30	$\eta = 4.15$, $\nu = .21$
Establishment exit rate	.10	.10	$\psi = .894$
Real interest rate	.045	.045	$\beta = .904$

(a) Table 1: calibration targets and parameters.

TABLE 2
MODEL DISTANCE TO DATA, RELATIVE TO PERFECT CREDIT TRANSITION

	Output	Investment Rate	TFP
5 years	.51	.23	.76
20 years	.70	.64	1.01

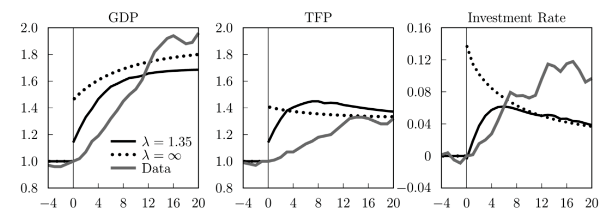
(b) Table 2: model distance to data relative to perfect credit transition.

6.3 Figure 3: Benchmark transition; Figure 4: Benchmark transition: capital and interest rate

Read: Figure 3 compares the benchmark financial-frictions transition, the perfect-credit transition, and the miracle-economy data. Figure 4 focuses on capital and the interest rate.

Economic message: Financial frictions slow reallocation. TFP and GDP rise gradually, investment starts low and rises, and capital accumulates slowly.

Detailed interpretation: Figure 3 is the main model-data comparison. The black solid line with financial frictions tracks the data better than the dotted perfect-credit transition because it reproduces delayed capital accumulation and a hump-shaped investment response. Figure 4 explains the macro price channel: capital demand by constrained entrepreneurs adjusts slowly, and the interest rate remains low when finance is tight.



IG. 3.—Benchmark transition: GDP, TFP, and investment rate. The black solid lines are the transitions from the benchmark exercise. For comp
transitional dynamics of a comparable neoclassical model are shown by the dotted lines, and the average of the economic miracles from Secti
resented by the gray lines (see App. B). The GDP and TFP series are normalized by their respective prereform values. Investment rates are sh
iations from their prereform levels. The horizontal axes show years, and year 0 corresponds to the reform date.

(a) Figure 3: benchmark transition in GDP, TFP, and investment.

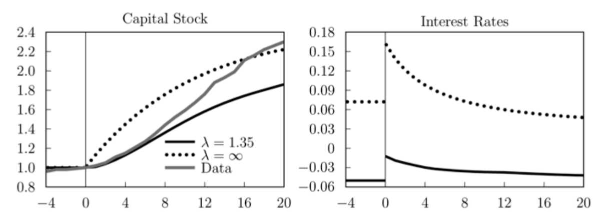


FIG. 4.—Benchmark transition: capital and interest rate. The black solid lines are the transitions from the benchmark exercise. Comparable neoclassical transition dynamics are shown by the dotted lines, and the evolution of the capital stock in the data, averaged across the miracle economies, is plotted by a gray solid line in the left panel (see App. B). The capital stock series are normalized by their respective prereform values. The horizontal

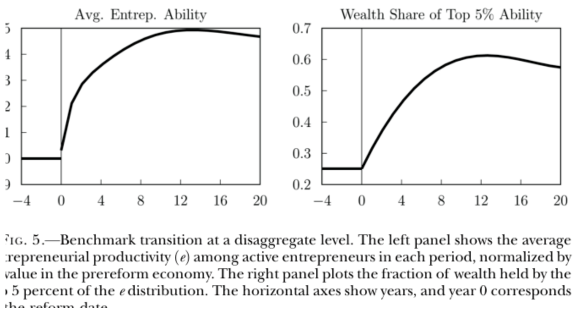
(b) Figure 4: benchmark transition in capital and interest rates.

6.4 Figure 5: Benchmark transition at a disaggregate level; Table 3: Model distance to data for reform and financial development

Read: Figure 5 shows average entrepreneurial ability among active entrepreneurs and wealth share of the top 5 percent of ability. Table 3 reports the model distance when reform and gradual financial development are combined.

Economic message: The model's micro mechanism is gradual reallocation toward high-productivity entrepreneurs. The top ability group accumulates wealth because collateral makes wealth necessary for expansion.

Detailed interpretation: Figure 5 is the clearest bridge from the aggregate transition to the micro mechanism. TFP grows because active entrepreneurs become more productive on average. Wealth concentration among talented entrepreneurs rises because they need collateral to enter and expand. Table 3 shows that allowing gradual financial development improves the fit further, especially for TFP.



(a) Figure 5: disaggregate transition in entrepreneurial ability and wealth concentration.

TABLE 3

MODEL DISTANCE TO DATA, RELATIVE TO PERFECT CREDIT TRANSITION FOR THE CASE OF REFORM AND FINANCIAL DEVELOPMENT

	Output	Investment Rate	TFP
5 years	.34	.28	.43
20 years	.47	.45	.59

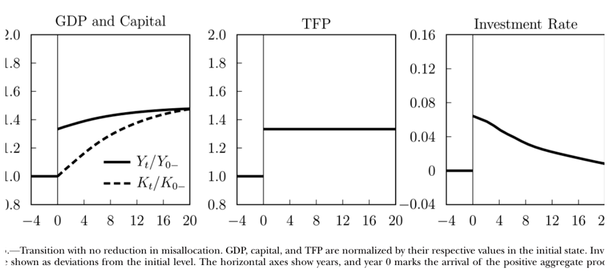
(b) Table 3: model distance for reform plus financial development.

6.5 Figure 6: Permanent technology shock; Figure 7: Financial development transition

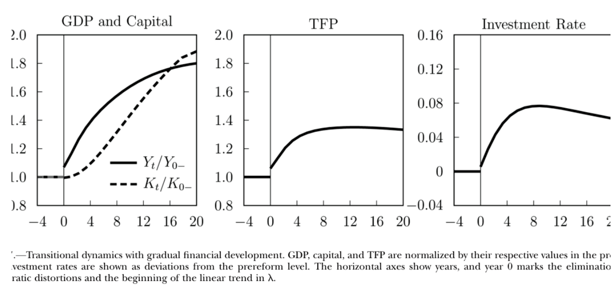
Read: Figure 6 shows a standard aggregate technology shock. Figure 7 shows a transition in which financial development improves gradually.

Economic message: A technology shock creates a fast transition and an investment-rate path inconsistent with the miracle data. Gradual financial development slows reallocation and improves the match to the data.

Detailed interpretation: The technology-shock comparison is a falsification of the standard neoclassical account. It generates high early investment and fast capital adjustment. The financial-development transition preserves the financial-frictions mechanism while allowing credit to deepen gradually, closer to what is observed in the data.



(a) Figure 6: transitional dynamics after a permanent technology shock.



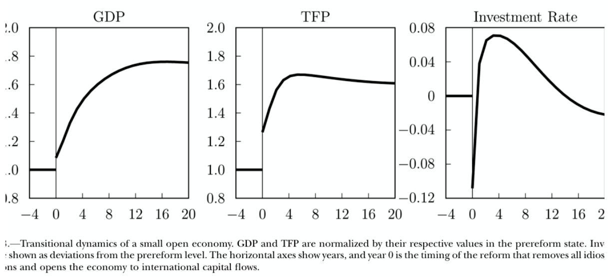
(b) Figure 7: transition with gradual financial development.

6.6 Figure 8: Reform in a small open economy; Figure 9: Micro-level evidence

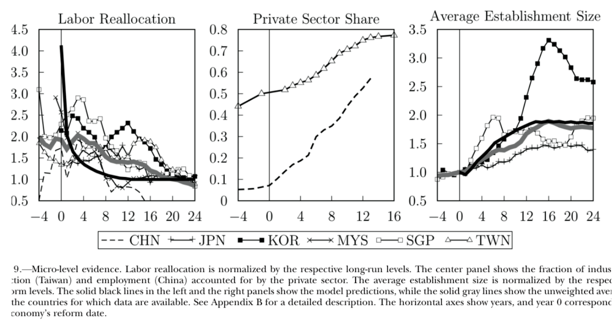
Read: Figure 8 shows an open-economy version with the world interest rate. Figure 9 shows micro evidence on labor reallocation, private-sector share, and average establishment size.

Economic message: Opening the capital account changes capital flows, but local financial frictions still shape domestic production. Figure 9 shows the micro patterns that the model predicts: reallocation and private-sector growth.

Detailed interpretation: Figure 8 clarifies that international capital flows do not eliminate domestic financial frictions. Local entrepreneurs still face collateral limits. Figure 9 supports the mechanism with evidence that labor moves, private-sector shares rise, and establishment size changes during miracle transitions.



(a) Figure 8: reform in a small open economy.



(b) Figure 9: micro-level evidence from miracle economies.

6.7 Appendix Figure C1: Alternative initial conditions; Appendix Figure C2: Alternative span of control

Read: These appendix figures vary initial conditions and the span-of-control parameter.

Economic message: The benchmark dynamics are robust to reasonable variation in initial capital/wealth and technology curvature.

Detailed interpretation: The appendix figures show that the qualitative mechanism is not a knife-edge artifact. Initial wealth and span of control affect the speed and magnitude of transition, but financial frictions still generate slow reallocation and gradual investment dynamics.

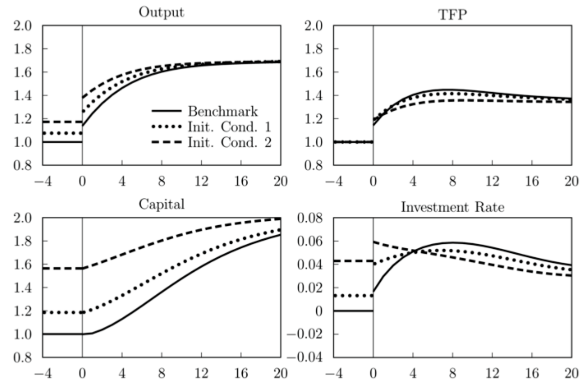
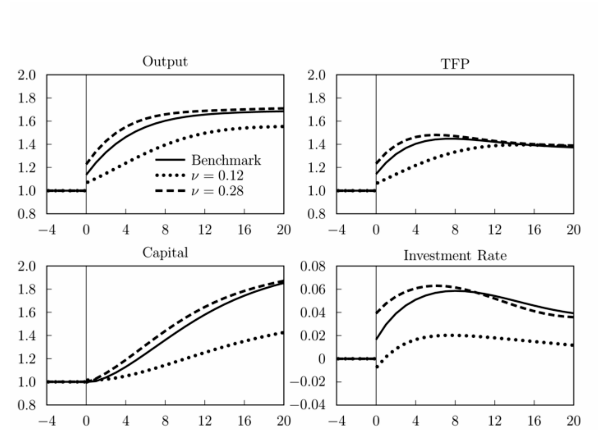


Fig. C1.—Transitions with alternative initial conditions

(a) Figure C1: transitions with alternative initial conditions.



(b) Figure C2: transitions with alternative span of control.

6.8 Appendix Figure C3: Alternative shock persistence

Read: Figure C3 varies the persistence of the entrepreneurial productivity shock.

Economic message: Higher persistence changes the value of entrepreneurship and the speed of reallocation, but the main transition mechanism remains.

Detailed interpretation: This robustness check is useful because entrepreneurial ability dynamics govern both entry and wealth accumulation. The model's qualitative result survives changes in persistence.

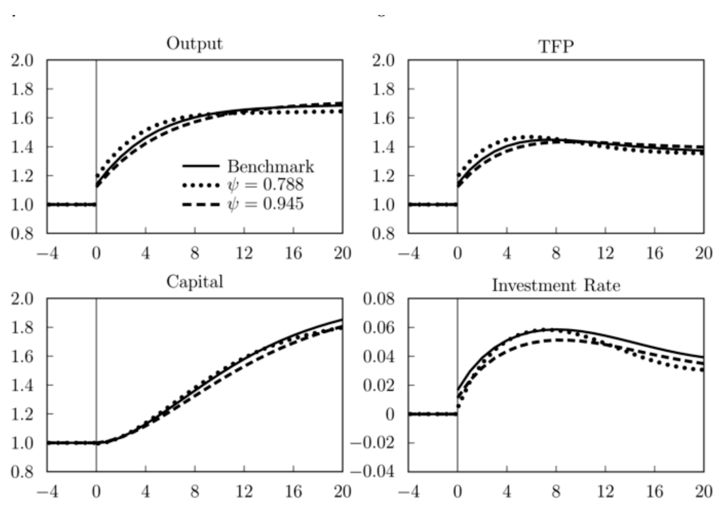


Fig. C3.—Transitions with alternative shock persistence

Figure C3: transitions with alternative shock persistence.

7 Short summary

- The paper explains why growth miracles exhibit slow output convergence, rising investment rates, and rising TFP.
- A reform removes idiosyncratic distortions that had misallocated resources.
- Financial frictions prevent immediate reallocation because talented entrepreneurs need wealth to borrow and expand.
- The model endogenizes TFP through changes in the allocation of capital and labor across heterogeneous entrepreneurs.
- Investment rates rise initially because productive constrained entrepreneurs accumulate wealth and gradually expand.
- Perfect-credit transitions are too fast and do not match the investment-rate pattern.
- Financial development arrives late in the data and strengthens the model's quantitative fit.
- Micro evidence on labor reallocation, private-sector growth, and establishment size supports the model's mechanism.

8 Conclusion

8.1 Core conclusion

Buera and Shin show that financial frictions can make history persistent. Even after a large reform removes distortions, the economy does not instantly reach an efficient allocation because the wealth distribution inherited from the distorted regime constrains who can become an entrepreneur and how large a scale they can operate.

8.2 Economic intuition

The central intuition is that efficient reallocation requires both talent and collateral. Reform reveals that some talented entrepreneurs should expand, but if they are initially poor, they cannot borrow enough. Subsidized low-productivity entrepreneurs exit slowly because their wealth cushions the transition. Over time, productive entrepreneurs save, accumulate wealth, enter, and expand. This gradual process generates rising TFP and rising investment rates.

8.3 Incremental contribution revisited

The paper's main contribution is to combine three literatures: misallocation, financial development, and transition dynamics. Static misallocation models explain low TFP levels. Financial-frictions models explain constrained entrepreneurship. Buera and Shin combine them to explain the shape and speed of development episodes after reform.

8.4 Why the paper matters

The paper changes how we think about reform. Removing distortions is not the end of development; it is the beginning of a transition. If financial markets are weak, reforms take time to translate into high productivity because productive firms must finance themselves internally.

8.5 Limitations

The model is stylized and uses reduced-form idiosyncratic distortions. It abstracts from many country-specific institutions, political constraints, sectoral transitions, human capital, and international linkages. The reform date is modeled as a discrete event, whereas actual reforms often unfold gradually. Still, the model matches broad facts that a standard neoclassical model misses.

8.6 Takeaway

The takeaway is that financial frictions turn past misallocation into future persistence. Growth miracles are slow not because reforms fail, but because productive entrepreneurs must build the collateral needed to make reforms effective.